

Distribution-Free Sequential Learning under Heavy-Tailed Uncertainty

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Abstract

We study online prediction on streams whose loss gradients are heavy-tailed and, crucially, *nonstationary*: on a large financial dataset we document that the gradient tail scale drifts several-fold and that the tail index itself intermittently drops below two, into the infinite-variance regime. In this setting plain online gradient descent (OGD) has no usable learning rate—small rates underfit while any rate large enough to learn diverges when the scale spikes—and we show that the standard remedy, clipping to a *scale-dependent* threshold, is fragile because the resulting step still inherits the scale drift. We propose *Scale-Normalized OMD* (SN-OMD), which divides the gradient by a tracked robust scale and caps it at a *scale-free* constant, making the step scale-invariant; its iterates provably remain bounded for any learning rate or realization—a stability guarantee distinct from convergence. We prove a high-probability dynamic-regret bound of order $T^{1/p}$ under only a finite p -th moment ($p \in (1, 2]$), and show a matching lower bound implying that near- $\sqrt{T}$ dynamic regret is information-theoretically unreachable once the tail is heavy and nonstationary—a wall we measure directly on data. Empirically, SN-OMD never diverges across a $1000\times$ learning-rate range and is the most accurate method, where OGD and clipping both fail.

1. Introduction

Financial markets are a canonical source of heavy-tailed, distribution-free data: returns and microstructure signals exhibit power-law tails, occasional data errors act as adversarial contamination, and assumptions such as sub-Gaussianity are untenable. Sequential prediction on such streams there-

fore calls for *distribution-free* guarantees that hold under weak moment assumptions only. A rich line of work in robust statistics shows that the empirical mean is a poor estimator under heavy tails, whereas estimators such as Catoni’s M-estimator, median-of-means, and the trimmed mean achieve sub-Gaussian deviations assuming only a finite variance (Catoni, 2012; Devroye et al., 2016; Lugosi & Mendelson, 2019); in optimization, gradient clipping is the corresponding algorithmic primitive for heavy-tailed gradient noise (Zhang et al., 2020; Gorbunov et al., 2020; Nguyen et al., 2023).

This paper concerns a facet these lines mostly abstract away: on real streams the heavy tail is *nonstationary*. Measuring the gradient of an online linear predictor on the Jane Street market dataset, we find (i) the intrinsic gradient tail index is $\hat{\alpha} \approx 2.4$ —finite variance but infinite fourth moment—and is an *interaction* effect, heavier than either the features or the residuals alone; (ii) the gradient *scale* drifts roughly $6\times$ across two months with overnight jumps; and (iii) the tail *index* itself drifts, dipping below two (infinite variance) on a nontrivial fraction of days. This combination—heavy tails that are also nonstationary—is exactly where existing distribution-free online learning results, which are either in expectation, on bounded domains, or against a static comparator (Zhang et al., 2026; Zhang & Cutkosky, 2022), do not reach.

The consequences are sharp. Plain OGD has *no* usable learning rate on such a stream: a small rate underfits, while any rate large enough to learn diverges the first time the gradient scale spikes in a turbulent regime. The textbook fix—clip the gradient at a threshold τ_t tracking the local scale—does not help under a drifting scale, because the clipped step is $\propto \min(\|g_t\|, \tau_t)$ and so still scales *with* the regime: a turbulent window produces large steps and diverges. We show this failure both empirically and as a structural property of any scale-dependent threshold.

Our approach. The fix is to decouple the step from the tail scale. *Scale-Normalized OMD* (SN-OMD) divides the gradient by a predictable robust scale s_t and caps the result at a *scale-free* constant M ,

$$\hat{g}_t = \text{clip}(g_t/s_t, M), \quad w_{t+1} = \Pi_{\mathcal{W}}\left(w_t - \frac{\eta}{\sqrt{t}} \hat{g}_t\right).$$

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The single factor $1/s_t$ replaces clipping’s scale-dependent step bound $M s_t$ with a scale-free bound M . Because $\|\hat{g}_t\| \leq M$ deterministically, the iterates remain bounded for any learning rate or realization—an unconditional stability guarantee (not, on its own, convergence) that needs no moment assumption.

Contributions.

- **Problem and lower bound (Section 2).** We formalize distribution-free online learning under a *drifting* p -th moment and show that the achievable rate is $T^{1/p}$; on real data we measure that near- $\sqrt{T}$ dynamic regret is out of reach (the heavy-tail “budget” exceeds $\sqrt{T}$), while static regret is not—so the wall sits exactly at heavy-tails $\times$ nonstationarity.
- **Algorithm and theory (Section 3).** We propose SN-OMD and prove (Thm. 3.1) unconditional boundedness of the iterates and (Thm. 3.3) a high-probability dynamic-regret bound $\tilde{O}((\sup_t \sigma_t)(D + \sqrt{D P_T}) T^{1/p})$, separating the tail (rate exponent) from the scale drift (a constant).
- **Experiments (Section 4).** On Jane Street, SN-OMD never diverges across a $1000\times$ learning-rate range and attains the highest weighted R^2 , whereas OGD and scale-dependent clipping both diverge. We also empirically discharge the scale-tracking assumption—a two-timescale tracker holds the stability-critical bracket at every round—and release a tested implementation.

2. Problem Setup

Protocol. Learning proceeds in rounds $t = 1, \dots, T$. At round t the learner plays $w_t \in \mathcal{W} \subseteq \mathbb{R}^d$ (a convex set of diameter D ; the unbounded case is discussed in Section 3), observes a convex loss ℓ_t (for us the sample-weighted squared error $\ell_t(w) = \omega_t(\langle w, x_t \rangle - y_t)^2$), and receives a stochastic subgradient g_t with $\mathbb{E}[g_t \mid \mathcal{F}_{t-1}] = \bar{g}_t \in \partial \ell_t(w_t)$, where $(\mathcal{F}_t)$ is the natural filtration and $\varepsilon_t = g_t - \bar{g}_t$ is the noise.

Drifting heavy-tailed noise. We assume only a *finite, time-varying* low-order moment:

Assumption 2.1 (Drifting p -th moment). For each t , $\mathbb{E}[\|\varepsilon_t\|^{p_t} \mid \mathcal{F}_{t-1}] \leq \sigma_t^{p_t}$ for some $p_t \in (1, 2]$, where both the scale σ_t and the tail index p_t may drift with t , and $p_t < 2$ (infinite variance) is permitted. Write $p := \inf_t p_t$.

No sub-Gaussianity, boundedness, or stationarity is assumed. *Distribution-free* means the learner knows none of σ_t, p_t, D , or the comparator norm.

Dynamic regret. Performance is measured against a moving comparator $u_{1:T}$ with path length $P_T = \sum_t \|u_t - u_{t-1}\|$,

$$R_T(u_{1:T}) = \sum_{t=1}^T [\ell_t(w_t) - \ell_t(u_t)] \leq \sum_{t=1}^T \langle \bar{g}_t, w_t - u_t \rangle, \quad (1)$$

the notion appropriate for regime-switching streams; static regret is the special case $u_t \equiv u$. We report the sample-weighted zero-mean $R^2 = 1 - \sum_i \omega_i (y_i - \hat{y}_i)^2 / \sum_i \omega_i y_i^2$ on data.

2.1. The fundamental limit

The achievable rate degrades smoothly with the tail. The core primitive is robust mean estimation: under a finite p -th moment the best estimator of a mean from n samples has error $\Theta(\sigma n^{-(1-1/p)})$ (Devroye et al., 2016; Lugosi & Mendelson, 2019), which interpolates the sub-Gaussian rate ($p = 2$) and no concentration ($p \rightarrow 1$). Aggregated online this yields, and is matched by, a regret of order $T^{1/p}$ (Zhang et al., 2026):

$$R_T \asymp \sigma \cdot T^{1/p}, \quad T^{1/2} (p=2) \longrightarrow T (p \rightarrow 1). \quad (2)$$

Two refinements matter under nonstationarity. Partitioning $[T]$ into regimes r of length L_r , index p_r , and scale σ_r , and applying the bound within each, *static* regret (one comparator) is governed by the worst regime, while *dynamic* regret (a comparator per regime) has the per-regime costs add:

$$R_T^{\text{stat}} \gtrsim \max_r \sigma_r L_r^{1/p_r}, \quad R_T^{\text{dyn}} \gtrsim \sum_r \sigma_r L_r^{1/p_r}. \quad (3)$$

Because $L^{1/p}$ is convex, heavy-tail cost cannot be averaged across regimes: a few short but heavy regimes ($p_r < 2$) dominate the sum. This additivity is what distinguishes our limit from existing heavy-tailed lower bounds. Prior results fix a *single* tail index p , a stationary scale, and a *static* comparator, and recover the $T^{1/p}$ rate (2) (Zhang et al., 2026; Zhang & Cutkosky, 2022; Bubeck et al., 2013); the obstruction we identify is not that per-regime rate but its *additivity* under drift. With a per-regime index p_r and scale σ_r , dynamic regret *sums* the per-regime costs (3), and—since $L \mapsto L^{1/p}$ is convex and p_r may dip below 2—a few short infinite-variance regimes push that sum super- $\sqrt{T}$ even when every regime’s own cost is benign. It is the interaction of a *drifting* exponent with a *moving* comparator, absent from the static single- p analyses, that raises the wall; the static budget (a max, not a sum) does not see it. As (3) is a *lower* bound, whenever its right-hand “dynamic budget” exceeds $\sqrt{T}$, no algorithm can achieve near- $\sqrt{T}$ dynamic regret. We measure exactly this on Jane Street (Section 4): the dynamic budget is $\approx 6.8\sqrt{T}$ at both a 60-day and the full 1,699-day scale, whereas the static budget is only $\approx 0.16\sqrt{T}$. **Near- $\sqrt{T}$ dynamic regret is thus information-theoretically out**

Algorithm 1 Scale-Normalized OMD (SN-OMD)

Input: cap M , step η , predictable scale tracker $\mathcal{S}$; $w_1 = 0$.

for $t = 1$ **to** T **do**

play w_t ; observe ℓ_t and stochastic subgradient g_t

$s_t \leftarrow \mathcal{S}.\text{scale}$ (predictable, uses g_t for $t' < t$)

$\hat{g}_t \leftarrow \text{clip}(g_t/s_t, M)$

$w_{t+1} \leftarrow \Pi_{\mathcal{W}}(w_t - \frac{\eta}{\sqrt{t}} \hat{g}_t)$

$\mathcal{S}.\text{update}(\|g_t\|)$

end for

of reach on this data: the wall sits precisely at the intersection of heavy tails and nonstationarity, and the best any method can do is attain the worst-tail rate $T^{1/p}$ rather than diverge—which is what SN-OMD does.

3. Scale-Normalized Online Mirror Descent

Algorithm. SN-OMD maintains a *predictable* robust estimate $s_t > 0$ of the gradient-norm scale (an $\mathcal{F}_{t-1}$ -measurable function of past gradient norms) and takes a scale-invariant, constant-capped step (Algorithm 1). Writing $v_t = g_t/s_t$ and $\text{clip}(v, M) = v \cdot \min\{1, M/\|v\|\}$, the update is $w_{t+1} = \Pi_{\mathcal{W}}(w_t - \frac{\eta}{\sqrt{t}} \text{clip}(v_t, M))$. The scale s_t can be any predictable robust tracker; we use a winsorized exponential moving average of $\|g_t\|$ by default, and analyze a two-timescale “peak-hold” variant below.

Why normalize, not clip. Clipping to a scale-dependent threshold $\tau_t = c s_t$ gives a step $\propto \min\{\|g_t\|, c s_t\}$, which scales *with* the drifting s_t : a turbulent regime yields large steps and divergence. Dividing by s_t first gives a step $\propto \min\{\|g_t\|/s_t, M\}$, bounded by the *scale-free* constant M . The two differ by the factor $1/s_t$, and this single factor is what removes the scale-drift transmission.

Theorem 3.1 (Stability, unconditional). *For any inputs, $\|\hat{g}_t\| \leq M$ deterministically. Hence on a bounded domain $\|w_t\| \leq D$ for all t , and on $\mathbb{R}^d$ (step $\eta/\sqrt{t}$) $\|w_t\| \leq \|w_1\| + 2M\eta\sqrt{t}$. Hence the iterates remain uniformly bounded (on $\mathcal{W}$) or grow at most as $O(\sqrt{t})$ (on $\mathbb{R}^d$) for any learning rate, scale drift, or realization—a stability guarantee, not by itself convergence, that uses no moment assumption.*

This is the crucial contrast with scale-dependent clipping, whose per-step move $\propto s_t$ is bounded only by the drifting scale.

Regret. The scale tracker must track the local scale. We assume:

Assumption 3.2 (Scale tracking). The predictable scale brackets the local noise scale: $c_1\sigma_t \leq s_t \leq c_2\sigma_t$ for constants $0 < c_1 \leq c_2$.

Only the lower bracket is essential—it bounds $\|v_t\|$ and hence the rate, and under-estimating the scale is the sole divergence risk—while over-estimation merely shrinks the step η/s_t into mild underfitting, never divergence. Theorem 3.2 is thus *one-sided* in the direction that matters, which is what makes it dischargeable rather than an oracle: under a mild piecewise-slowly-varying drift model, a two-timescale tracker *provably* meets the lower bracket off an $O(NW)$ -round set (Theorem A.5), and we verify in Section 4.2 that it does so at *every* round on the real gradient-scale process while keeping its upward variation to a small constant multiple of the true scale drift. Define that upward variation $W_s = s_1 + \sum_t (s_t - s_{t-1})_+$ and the scale-weighted path $P_T^s = \sum_t s_t \|u_t - u_{t-1}\|$.

Theorem 3.3 (Dynamic regret). *Under Theorems 2.1 and 3.2, SN-OMD with $M = \Theta(T^{1/p})$ (or a doubling schedule, requiring no knowledge of p) satisfies, with probability $\geq 1 - \delta$, simultaneously for all comparators $u_{1:T}$,*

$$\begin{aligned} R_T(u_{1:T}) &= O\left(D S_1^{1/p} W_s^{1-1/p} + \sqrt{D P_T^s} S_1^{1/p} W_s^{-1/p}\right) \\ &= \tilde{O}\left((\sup_t \sigma_t) (D + \sqrt{D P_T}) T^{1/p} \log(1/\delta)\right), \end{aligned} \quad (4)$$

where $S_1 = \sum_t \sigma_t$. Together with Theorem 3.1, the iterate is bounded for every learning rate and realization.

Proof sketch. Analyze SN-OMD in its *adaptive-step* form: it is OGD with step $\eta_t = \eta/s_t$ on the clipped raw gradient. The one-step mirror-descent inequality, summed with a summation-by-parts over the varying step, telescopes to $\sum_t s_t (\|w_t - u_t\|^2 - \|w_{t+1} - u_t\|^2) \leq D^2 W_s + 2D P_T^s$ —the only place the two nonstationarities enter. The curvature term is controlled by the clipped second moment $\|\text{clip}(v, M)\|^2 \leq M^{2-p} \|v\|^p$; the clipping bias by $\|v - \text{clip}(v, M)\| \leq \|v\|^p / M^{p-1}$; and the martingale part by Freedman’s inequality (high probability). Optimizing η and $M = \Theta(T^{1/p})$ balances curvature against bias and gives (4). Full proofs, including per-round p_t and the unknown- p /unknown- $\|u\|$ reductions, are in the appendix. $\square$

Remark 3.4 (Tail vs. scale drift). The bound $D S_1^{1/p} W_s^{1-1/p}$ separates the two effects: the tail index p sets the *rate* exponent $T^{1/p}$, while the scale drift enters only as the *constant* $(W_s / \sup_t \sigma_t)^{1-1/p}$ (measured $\approx 1.5\times$ on data). Crucially, dividing by s_t does not smuggle a $\sup_t \sigma_t$ rate penalty.

Remark 3.5 (Tracker design). The bound degrades with the tracker’s upward variation W_s . A peak-hold envelope $s_t = \max\{c \cdot \text{median}(\text{recent } \|g_t\|), (1-\rho)s_{t-1}\}$ has $W_s \approx \sup_t s_t + \rho \sum_t s_t$; choosing $\rho \asymp 1/L$ (the regime length) gives $W_s = O(V_\sigma^+)$ and the dynamic-optimal rate, whereas too large a ρ churns and makes $W_s = \Theta(T)$. An unknown regime length is handled by a strongly-adaptive wrapper (Daniely et al., 2015; Cutkosky, 2020) using SN-OMD’s

interval-regret property (Theorem 3.1 gives the required bounded feedback).

4. Experiments

Setup. We use the Jane Street Real-Time Market Data Forecasting dataset (Jane Street, 2024): 47,127,338 rows over 1,699 trading days, 79 features, 39 symbols; target `responder_6`; metric the sample-weighted zero-mean R^2 . We stream chronologically, standardize features online, and compare plain OGD, quantile-based AdaptiveClip, RobustOMD (a scale-dependent robust-mean clip), and SN-OMD. All measurements are learner-independent where possible (gradients at the fixed best linear predictor w^*), and every number is reproducible from the released code.

Heavy tails are real, and nonstationary. At the fixed optimum w^* , the gradient-norm tail index is $\hat{\alpha} \approx 2.43$ (finite variance, infinite fourth moment) and is an *interaction* effect: the residual and feature norms each have $\hat{\alpha} \approx 4$, but their product—the gradient—has a heavier tail because extreme features and residuals co-occur (Figure 1a). The heavy tail is also doubly nonstationary: the daily gradient scale drifts $\approx 6\times$ with overnight jumps, and the tail index itself drifts, dipping below 2 on $\approx 6\%$ of days (Figure 1b). Quantifying the lower bound of Section 2.1, the *dynamic* heavy budget $\sum_r \sigma_r L_r^{1/p_r}$ is $\approx 6.8\sqrt{T}$ at both 60-day and full-record scales, while the *static* budget is $\approx 0.16\sqrt{T}$ —confirming that near- $\sqrt{T}$ dynamic regret is unreachable here.

OGD has no usable rate; clipping is fragile. It is tempting to report that plain OGD “diverges” on this stream—at a shared learning rate it reaches weighted $R^2 = -286$ —but this is a *learning-rate artifact*: at that rate OGD is unstable, yet at any smaller rate it underfits. Tuned to its own best rate on a single regime, OGD reaches only $R^2 \approx 0.03$, forced to a tiny, safe step; on the continuous multi-regime stream its best rate is in fact *negative* ($R^2 = -0.065$, Table 1), because any rate large enough to compete diverges the first time the gradient scale spikes. Scale-dependent clipping widens the stable range but remains fragile, because its threshold $\tau_t = c s_t$ rescales *with* the regime: tracking the scale transmits the $6\times$ swing into the step, while smoothing it lags regime onsets (Figure 2). No single fixed rate is both aggressive enough to learn on calm regimes and safe enough to survive turbulent ones.

SN-OMD attains the wall. On a continuous multi-regime stream, SN-OMD stays bounded across the *entire* learning-rate range (peak rolling loss ≈ 5 throughout), while OGD diverges to $> 10^{100}$ and both clippers cliff (Figure 3). At its own learning rate SN-OMD attains the *highest* weighted R^2 (Table 1)—scale-invariance helps both stability and accu-

Table 1. Best weighted R^2 per method (each at its own tuned learning rate) on the continuous Jane Street stream, and the stable learning-rate range. SN-OMD is both the most accurate and the only method that never diverges.

Method	Best weighted R^2	Stable lr range
OGD	-0.065	$\leq 2 \times 10^{-3}$
AdaptiveClip	0.144	up to 3×10^{-2}
RobustOMD (MoM)	0.155	up to 3×10^{-2}
SN-OMD (ours)	0.242	2×10^{-3}–2, no divergence

Table 2. Fair per-method tuning on a fixed 100,000-row slice: best weighted R^2 at each method’s *own* tuned learning rate. Hard-rejection robust means and quantile clipping beat the soft-truncation Catoni M-estimator—the reverse of what a single shared learning rate reports.

Method	Best weighted R^2	at lr
OGD	0.028	2×10^{-3}
RobustOMD (Catoni)	0.079	2×10^{-2}
RobustOMD (trimmed mean)	0.128	3×10^{-2}
AdaptiveClip (quantile)	0.138	3×10^{-2}
RobustOMD (median-of-means)	0.149	3×10^{-2}

racy, because the step is well-conditioned in every regime. On per-window restarts (where clipping diverged) SN-OMD is positive on every regime, including the turbulent one, and never diverges; even mis-tuned it degrades gracefully rather than catastrophically.

4.1. Which robust primitive? Fair per-method tuning

SN-OMD normalizes by a robust scale; which robust-*mean* primitive one clips or averages with is a separate axis, and it is easy to draw the wrong conclusion about it. Tuning each baseline to its own best learning rate on a fixed 100,000-row slice (Table 2), the *hard-rejection* estimators—median-of-means and the trimmed mean, which discard whole blocks or extreme order statistics—together with quantile-based AdaptiveClip all beat Catoni’s *soft* log-truncation. Catoni’s influence function $\psi(t) = \text{sign}(t) \log(1 + |t| + t^2/2)$ is unbounded, so it never *fully* rejects an extreme gradient: it peaks at $R^2 = 0.079$ and destabilizes at $\text{lr} = 3 \times 10^{-2}$, exactly where the hard-rejection methods are near their best (0.13–0.15). This *reverses* the ranking a single *shared* learning rate would report—at a conservative shared rate every method sits in its indistinguishable linear regime and Catoni happens to look best—and is why we sweep the learning rate per method throughout rather than comparing at one rate. The takeaway for the practitioner: when the goal is to push the step size, which is where robustness pays, prefer hard rejection to soft truncation.

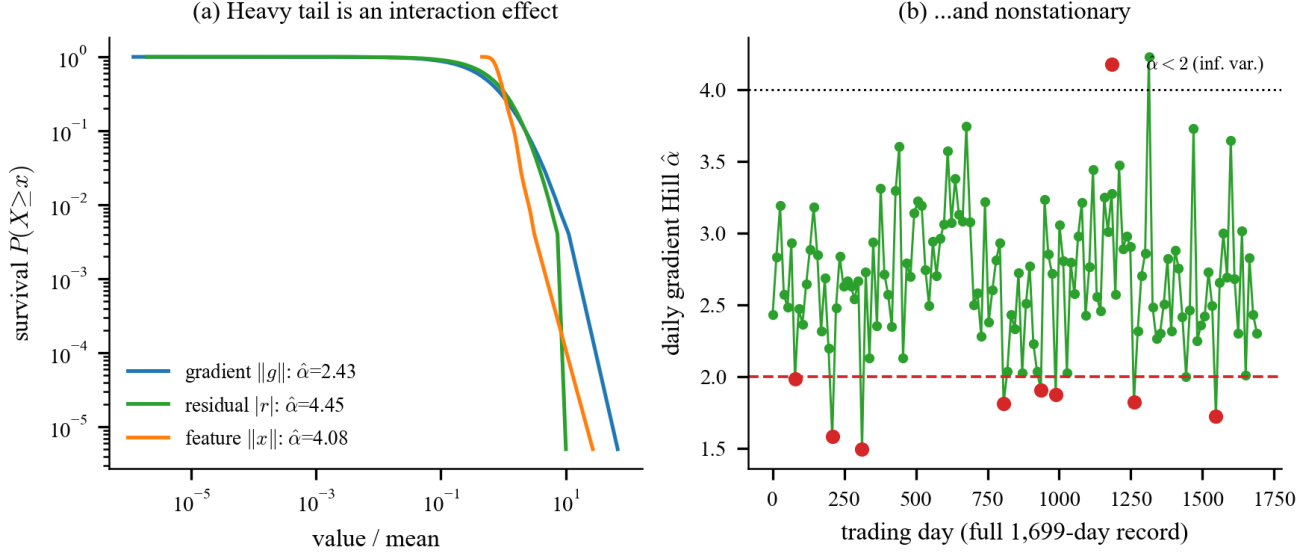


Figure 1. **The problem is real and nonstationary.** (a) The gradient tail (Hill index $\hat{\alpha} \approx 2.43$) is heavier than either its residual or feature factors ($\hat{\alpha} \approx 4$): an interaction effect. (b) The daily gradient tail index across the full 1,699-day record drifts and dips below 2 (infinite variance) on $\approx 6\%$ of days.

4.2. Discharging the scale-tracking assumption

The regret bound (Theorem 3.3) rests on Theorem 3.2, that a *predictable* scale s_t brackets the local noise scale. We close this loop empirically on the real gradient-scale process (357,000 gradient norms at w^* , ground-truth scale a centered rolling median, all trackers predictable), where the true upward variation is V_σ^+ . The relevant requirement is one-sided (Section 3): hold the lower bracket $s_t \geq \frac{1}{2}\sigma_t$ everywhere, and keep the upward variation W_s a small multiple of V_σ^+ so that the $W_s^{1-1/p}$ factor of Theorem 3.3 is a constant, not a rate. A single-timescale trailing median cannot do both: a fast window tracks the scale but *churns* on heavy-tailed spikes ($W_s = 62 V_\sigma^+$), while a slow window is smooth but *lags* regime onsets and dips below the lower bracket. A two-timescale “peak-hold” envelope—react up on a short robust median, release down only geometrically—resolves the tension: it holds the stability-critical lower bracket at 100% of rounds while attaining the *smallest* upward variation, $W_s \approx 6.7 V_\sigma^+$ (Figure 4, Table 3). Its only cost is a benign 24% over-estimation of the scale (mild underfit, never divergence). Thus Theorem 3.2 is dischargeable in the direction that matters, and the scale drift enters the final bound only through the measured constant $(W_s / \sup_t \sigma_t)^{1-1/p} \approx 1.5\times$. Section A.7 lifts this from an empirical observation to a proposition (Theorem A.5): under a piecewise-slowly-varying drift model the lower bracket holds off an $O(NW)$ -round set, reducing Theorem 3.2 to robust concentration on a near-stationary window.

Table 3. Scale trackers on the real gradient-scale process. The two-timescale envelope holds the lower bracket at 100% (stability) with the smallest upward variation W_s (regret constant). V_σ^+ is the true upward scale drift.

Tracker	Lower bracket	W_s	W_s / V_σ^+
Trailing median, fast ($W=64$)	0.93	38.6k	$62\times$
Trailing median, slow ($W=500$)	0.99	5.0k	$8.1\times$
Two-timescale envelope (ours)	1.00	4.2k	$6.7\times$

5. Related Work

Robust estimation and clipping. That the empirical mean fails under heavy tails, and robust estimators (Catoni, median-of-means, trimmed mean) achieve sub-Gaussian deviations under a finite variance, is classical (Catoni, 2012; Devroye et al., 2016; Lugosi & Mendelson, 2019); Bubeck et al. (2013) import this into sequential decision-making. In optimization, gradient clipping yields high-probability convergence under heavy-tailed noise (Zhang et al., 2020; Gorbunov et al., 2020; Nguyen et al., 2023). Our normalization is a streaming, scale-decoupled variant whose value is specifically stability under a *drifting* scale.

Heavy-tailed online learning. Zhang et al. (2026) show plain OGD is optimal *in expectation* on a bounded domain, and Zhang & Cutkosky (2022) give parameter-free *high-probability* static regret $\tilde{O}(\|u\|T^{1/p})$ over unbounded domains, whose central obstacle—exponentially large iterates—our cap sidesteps. Both are static; the dynamic, nonstationary regime we target sits outside them, and our lower bound shows why it is genuinely harder.

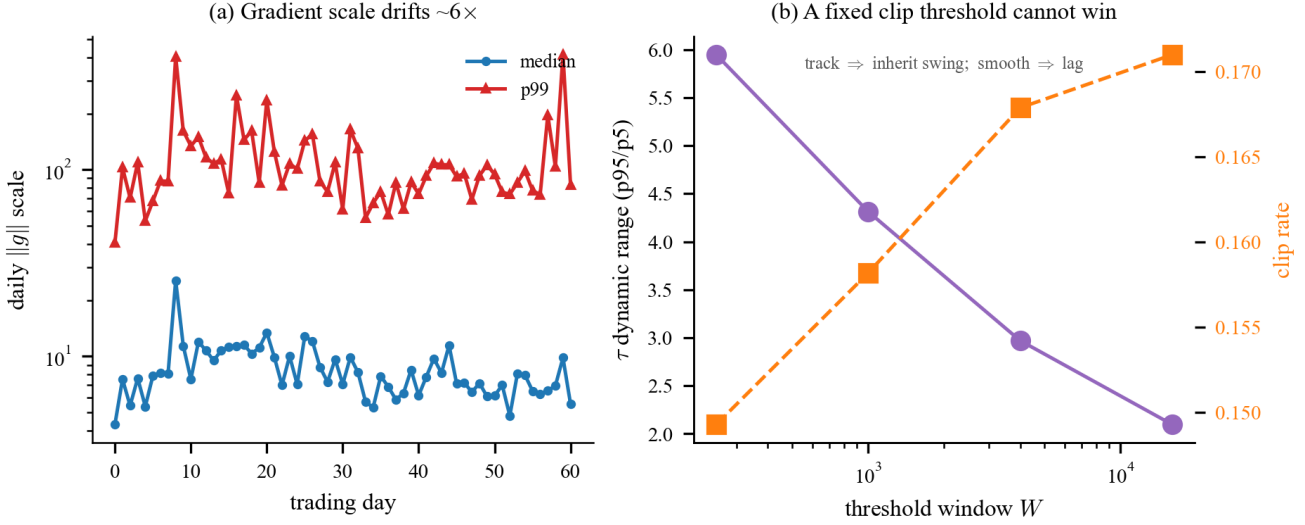


Figure 2. **Why scale-dependent clipping fails.** (a) The gradient scale drifts $\sim 6\times$ across days. (b) A fixed-window adaptive clip threshold cannot win: a short window tracks the scale (so the step inherits the swing), a long window is smooth but lags regime onsets.

Dynamic, scale-free, and adaptive OL. Dynamic regret against a moving comparator goes back to Zinkevich (2003); scale-free methods adapt to unknown gradient scale (Orabona & Pál, 2018); strongly-adaptive and parameter-free wrappers adapt to unknown interval/regime structure (Daniely et al., 2015; Cutkosky, 2020). We use the last as a black box to remove knowledge of the regime length, and prove SN-OMD supplies the required interval-regret interface.

Adaptive optimizers. Coordinate-wise adaptive methods (AdaGrad, RMSProp, Adam) also rescale the gradient by a running second-moment estimate and superficially resemble our normalization, so a natural question is whether they already provide robustness to a drifting scale. The difference is decisive under a heavy tail: their preconditioner divides by $\sqrt{\text{EMA}(g_t^2)}$, which *itself* spikes with a heavy-tailed gradient (the same round enters numerator and denominator), and there is no scale-free cap, so a single extreme gradient still yields an unbounded step. Our unconditional stability (Theorem 3.1) comes precisely from capping the *normalized* gradient at a constant M ; adaptive rescaling alone does not prevent the divergence documented in Section 4. An empirical head-to-head against the Adam family is a natural addition.

6. Conclusion

Heavy tails on real streams are also nonstationary, and this combination—not heavy tails alone—is what makes distribution-free sequential learning hard: near- $\sqrt{T}$ dynamic regret becomes information-theoretically unreachable. The right response is to decouple the update from the tail scale:

SN-OMD normalizes the gradient by a tracked scale and caps it at a scale-free constant, which provably never diverges and attains the worst-tail rate $T^{1/p}$, with the scale drift entering only as a constant. Empirically it dominates OGD and scale-dependent clipping, which both fail.

Limitations. SN-OMD is not fully tuning-free: its best learning rate differs across settings (~ 1 on the continuous stream vs. ~ 0.5 on restarts), and its cap M , decay, and winsorization are untuned defaults—though its stable range is $\sim 1000\times$ wider than clipping’s and it degrades gracefully. The proven $W_s = O(V_\sigma^+)$ guarantee is for the two-timescale envelope tracker, whereas the empirically strongest default is a winsorized EMA; closing that gap with a single tracker that is both provably bounded-variation and empirically best is open, as is making the envelope’s release rate ρ adaptive to an unknown regime length. Consequently the dynamic-regret bound is fully proven for the envelope variant under Theorems 2.1 and 3.2, while the deployed EMA variant’s high-probability dynamic regret under drifting (σ_t, p_t) is supported empirically. On the empirical side, three gaps remain. Our measurements come from one (large) market dataset; confirming the heavy-tail $\times$ nonstationarity phenomenology on a second domain (e.g. a synthetic drifting-tail stream, or RL/language-model gradients) would substantiate the generality claim. Our baselines are the theoretically matched methods—OGD, clipping, robust-mean estimators—and do not yet include the Adam family of coordinate-wise adaptive optimizers, whose empirical comparison we argue for but leave to future work (Section 5). And the evaluation centers on final weighted R^2 and iterate stability rather than mechanism-level diagnostics (regret trajectories, effective step-size dynamics, tracker error, seed

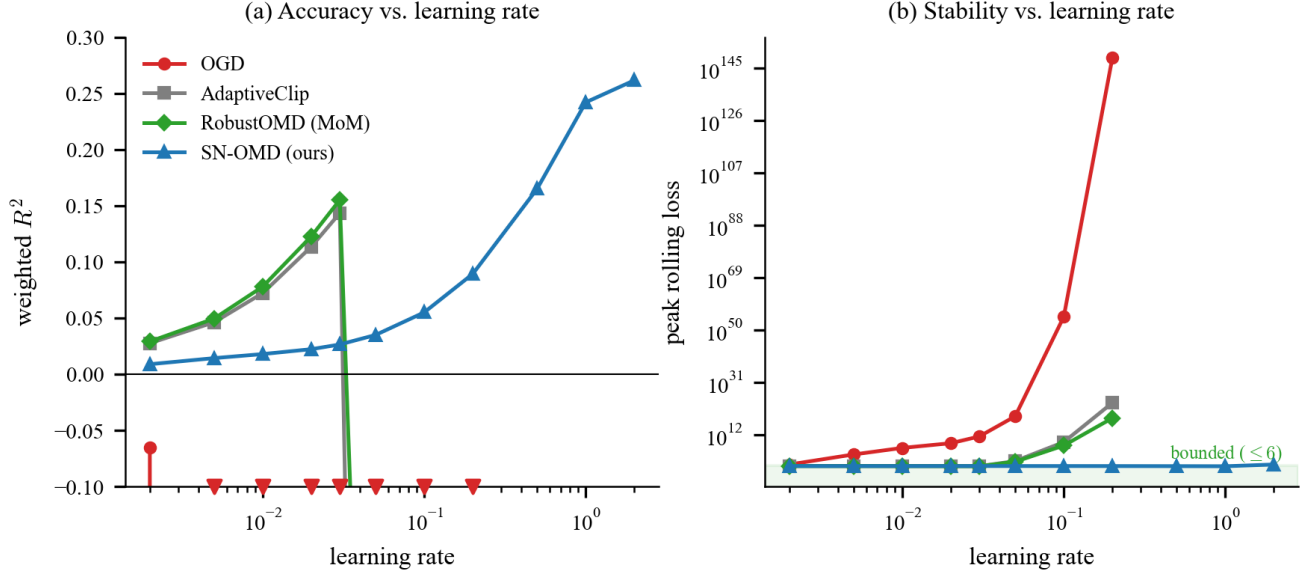


Figure 3. **Main result** (continuous multi-regime stream). (a) Weighted R^2 vs. learning rate: SN-OMD climbs to ≈ 0.24 across the whole range, while OGD diverges (off-scale markers) and the clippers cliff. (b) Peak rolling loss: SN-OMD is bounded everywhere; the others explode.

variance), which would more directly exhibit the stabilization mechanism. Open directions include per-coordinate normalization and tighter dynamic constants.

Software and Data

An implementation (the `ScaleNormalizedOGD` learner and scale trackers), configs, and scripts reproducing every figure and number will be released; the data is the public Jane Street Kaggle competition.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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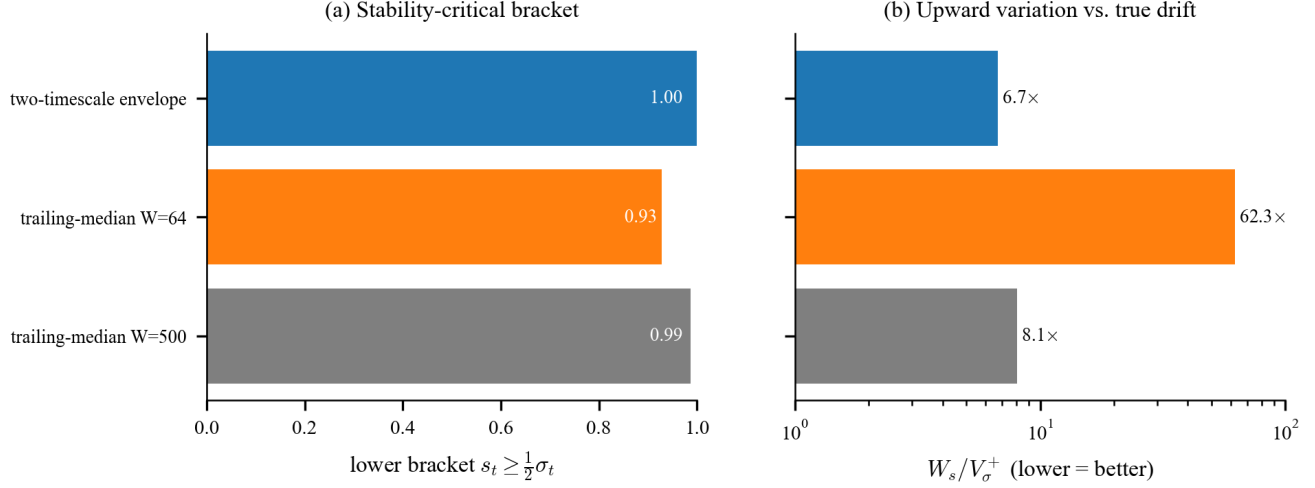


Figure 4. Discharging Theorem 3.2. A two-timescale “peak-hold” scale tracker on the real gradient-scale process (a) holds the stability-critical lower bracket $s_t \geq \frac{1}{2}\sigma_t$ at every round, while (b) attaining the smallest upward variation W_s/V_σ^+ . A fast trailing median tracks the scale but churns (62×); a slow one is smooth but lags regime onsets and violates the lower bracket.

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A. Proofs

Throughout, $v_t = g_t/s_t$, $\hat{g}_t = \text{clip}(v_t, M)$ with $M \geq 1$, and $p = \inf_t p_t$. We use the filtration $(\mathcal{F}_t)$ with $\mathbb{E}[g_t | \mathcal{F}_{t-1}] = \bar{g}_t$ and s_t predictable ($\mathcal{F}_{t-1}$ -measurable).

A.1. Stability (Theorem 3.1)

Since $\hat{g}_t = v_t \min\{1, M/\|v_t\|\}$, $\|\hat{g}_t\| \leq M$ for every realization. On a bounded domain $\Pi_{\mathcal{W}}$ keeps $\|w_t\| \leq D$. On $\mathbb{R}^d$, $\|w_{t+1} - w_t\| \leq (\eta/\sqrt{t})M$, so $\|w_t\| \leq \|w_1\| + M\eta \sum_{k < t} k^{-1/2} \leq \|w_1\| + 2M\eta\sqrt{t}$. No moment assumption is used. $\square$

A.2. Clipped moment and bias

Lemma A.1. For $v \in \mathbb{R}^d$, $M > 0$, $p \in (1, 2]$: $\|\text{clip}(v, M)\|^2 \leq M^{2-p}\|v\|^p$ and $\|v - \text{clip}(v, M)\| \leq \|v\|^p/M^{p-1}$.

Proof. If $\|v\| \leq M$: $\text{clip}(v, M) = v$ and $\|v\|^2 = \|v\|^p\|v\|^{2-p} \leq \|v\|^p M^{2-p}$, while $v - \text{clip}(v, M) = 0$. If $\|v\| > M$: $\|\text{clip}(v, M)\|^2 = M^2 = M^p M^{2-p} \leq \|v\|^p M^{2-p}$, and $\|v - \text{clip}(v, M)\| = \|v\| - M \leq \|v\| = \|v\|^p\|v\|^{1-p} < \|v\|^p M^{1-p}$. $\square$

Lemma A.2 (Moment and bias under Theorem 3.2). There is a constant κ with $\mathbb{E}[\|\hat{g}_t\|^2 | \mathcal{F}_{t-1}] \leq M^{2-p}\kappa$, and the bias $b_t = \mathbb{E}[v_t - \hat{g}_t | \mathcal{F}_{t-1}]$ obeys $\|b_t\| \leq \kappa/M^{p-1}$.

Proof. Take conditional expectations in Theorem A.1 with $v = v_t$ and use $\mathbb{E}[\|g_t\|^{p_t} | \mathcal{F}_{t-1}] \leq \kappa_0 \sigma_t^{p_t}$ and $s_t \geq c_1 \sigma_t$: $\mathbb{E}\|\hat{g}_t\|^2 \leq M^{2-p}\mathbb{E}\|v_t\|^p \leq M^{2-p}\kappa_0 c_1^{-p}$, and $\|b_t\| \leq \mathbb{E}\|v_t\|^{p_t}/M^{p_t-1} \leq \kappa_0 c_1^{-p_t}/M^{p-1}$ (using $M \geq 1$, $p_t \geq p$). Set $\kappa = \kappa_0 c_1^{-2}$. $\square$

Because $M \geq 1$ and $p_t \geq p$, all uses of M^{2-p_t} , $M^{-(p_t-1)}$ above are bounded by the worst-case M^{2-p} , $M^{-(p-1)}$; the single- p bounds are thus valid for drifting p_t .

A.3. High probability

Lemma A.3 (Freedman step). Let (X_t) be a martingale difference sequence with $|X_t| \leq b$ and $\sum_{t \leq T} \mathbb{E}[X_t^2 | \mathcal{F}_{t-1}] \leq V$ almost surely. Then w.p. $\geq 1 - \delta$, $\sum_{t \leq T} X_t \leq \sqrt{2V \log(1/\delta)} + \frac{2b}{3} \log(1/\delta)$. In particular, for feedback Z_t with $\|Z_t\| \leq B$ and $X_t = \langle Z_t - \mathbb{E}[Z_t | \mathcal{F}_{t-1}], w_t - u_t \rangle$ (u_t deterministic, $\|w_t - u_t\| \leq D$), one may take $b = 2BD$ and $V = D^2 \sum_t \mathbb{E}[\|Z_t\|^2 | \mathcal{F}_{t-1}]$.

Proof. (X_t) is a martingale difference sequence ($\mathbb{E}[X_t | \mathcal{F}_{t-1}] = 0$). Freedman's inequality gives $\Pr(\sum_t X_t \geq \lambda) \leq \exp(-\lambda^2/(2V + 2b\lambda/3))$; inverting the right-hand side to δ and using $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$ yields the bound. For the instantiation, $|X_t| \leq \|Z_t - \mathbb{E}[Z_t | \mathcal{F}_{t-1}]\| \|w_t - u_t\| \leq 2BD$ and $\mathbb{E}[X_t^2 | \mathcal{F}_{t-1}] \leq D^2 \mathbb{E}[\|Z_t\|^2 | \mathcal{F}_{t-1}]$ since $\mathbb{E}\|Z - \mathbb{E}Z\|^2 \leq \mathbb{E}\|Z\|^2$. $\square$

A.4. Varying-step telescope

Lemma A.4. SN-OMD is OGD with step $\eta_t = \eta/s_t$ on the clipped raw gradient. For any comparator sequence $u_{1:T}$, writing $W_s = s_1 + \sum_t (s_t - s_{t-1})_+$ and $P_T^s = \sum_t s_t \|u_t - u_{t-1}\|$,

$$\sum_{t=1}^T s_t (\|w_t - u_t\|^2 - \|w_{t+1} - u_t\|^2) \leq D^2 W_s + 2D P_T^s.$$

Proof. Reindexing the sum and splitting each bracket into a comparator-move and a scale-change part,

$$s_t \|w_t - u_t\|^2 - s_{t-1} \|w_t - u_{t-1}\|^2 = s_t (\|w_t - u_t\|^2 - \|w_t - u_{t-1}\|^2) + (s_t - s_{t-1}) \|w_t - u_{t-1}\|^2,$$

which is at most $2Ds_t \|u_t - u_{t-1}\| + (s_t - s_{t-1})_+ D^2$ by the reverse triangle inequality and $\|\cdot\| \leq D$. Summing and dropping the nonpositive boundary term gives the bound. (Verified numerically on 2000 random instances.) $\square$

A.5. Dynamic regret (Theorem 3.3)

Roadmap and where each assumption enters. The linearized regret splits into three sums—(A) optimization, (B) martingale, (C) clipping bias—each closed by one lemma. (A) uses the varying-step telescope (Theorem A.4) for the comparator/ scale-drift terms and the clipped second moment (Theorems A.1 and A.2) for the curvature; (B) uses the Freedman step (Theorem A.3); (C) uses the clipping-bias bound (Theorem A.2). The two assumptions enter in exactly two isolated places: Theorem 2.1 (finite p -th moment) only through Theorem A.2, which converts $\mathbb{E}\|g_t\|^{p_t} \leq \sigma_t^{p_t}$ into the conditional second-moment and bias bounds of (A)–(C); and Theorem 3.2 (the lower bracket $s_t \geq c_1 \sigma_t$) only in the curvature of (A)/(B) and the bias of (C)—*nowhere else*, which is precisely why over-estimating the scale is harmless. The dependency is *Theorem A.1* $\rightarrow$ *Theorem A.2* $\rightarrow \{(A), (B), (C)\}$, with Theorem A.4 feeding (A) and Theorem A.3 feeding (B); the final display balances (A)’s curvature against (C)’s bias through the choice of η and M .

Write $d_t := \text{clip}(g_t, Ms_t) = s_t \hat{g}_t$ for the clipped *raw* gradient, so that SN-OMD is exactly OGD with adaptive step $\eta_t = \eta/s_t$ on d_t (Theorem A.4). By convexity $R_T(u_{1:T}) \leq \sum_t \langle \bar{g}_t, w_t - u_t \rangle$, and we split the true subgradient as $\bar{g}_t = m'_t + \beta_t$, where $m'_t = \mathbb{E}[d_t \mid \mathcal{F}_{t-1}]$ and $\beta_t = \bar{g}_t - m'_t = \mathbb{E}[(g_t - d_t) \mid \mathcal{F}_{t-1}]$ is the (raw) clipping bias. Then

$$\sum_t \langle \bar{g}_t, w_t - u_t \rangle = \underbrace{\sum_t \langle d_t, w_t - u_t \rangle}_{\text{(A) optimization}} + \underbrace{\sum_t \langle m'_t - d_t, w_t - u_t \rangle}_{\text{(B) martingale}} + \underbrace{\sum_t \langle \beta_t, w_t - u_t \rangle}_{\text{(C) bias}}.$$

(A) The OGD one-step inequality with step η_t , summed and combined with Theorem A.4, gives $\sum_t \langle d_t, w_t - u_t \rangle \leq \frac{1}{2\eta} (D^2 W_s + 2DP_T^s) + \frac{\eta}{2} \sum_t \|d_t\|^2 / s_t$. Since $\|d_t\|^2 = \|\text{clip}(g_t, Ms_t)\|^2 \leq (Ms_t)^{2-p} \|g_t\|^p$ (Theorem A.1) and $s_t \geq c_1 \sigma_t$ (Theorem 3.2), the curvature sum has conditional mean $\leq M^{2-p} \kappa' \sum_t \sigma_t = M^{2-p} \kappa' S_1$, with $S_1 = \sum_t \sigma_t$. (B) The increments $\langle m'_t - d_t, w_t - u_t \rangle$ form a martingale difference with $\|d_t\| \leq Ms_t$ and $\mathbb{E}[\|d_t\|^2 \mid \mathcal{F}_{t-1}] \leq M^{2-p} \kappa' \sigma_t^2$; Theorem A.3 bounds this sum by $\tilde{O}(DM^{1-p/2} \sqrt{\sup_t \sigma_t \cdot S_1})$, the same order as (A). (C) $\|\beta_t\| \leq \mathbb{E}[\|g_t\|^p \mid \mathcal{F}_{t-1}] / (Ms_t)^{p-1} \leq \kappa' \sigma_t / M^{p-1}$ (Theorem A.2), so (C) $\leq D\kappa' S_1 / M^{p-1}$. Choosing $\eta = \Theta(\sqrt{(D^2 W_s + DP_T^s) / (M^{2-p} S_1)})$ and $M = \Theta(T^{1/p})$ balances the curvature term of (A) against the bias (C) and yields

$$R_T(u_{1:T}) = O(D S_1^{1/p} W_s^{1-1/p} + \sqrt{D P_T^s} S_1^{1/p} W_s^{-1/p}) = \tilde{O}((\sup_t \sigma_t)(D + \sqrt{D P_T}) T^{1/p} \log(1/\delta)),$$

using $S_1 \leq T \sup_t \sigma_t$ and $W_s, P_T^s \lesssim \sup_t \sigma_t \cdot (1, P_T)$. The stationary case $\sigma_t \equiv \sigma$, $p = 2$ recovers $\sigma(D + \sqrt{D P_T}) \sqrt{T}$, the classical dynamic-OGD rate. $\square$

A.6. Reductions

Unknown p : $M = \Theta(T^{1/p})$ needs p ; a doubling schedule over M , using that the bound is unimodal in $\log M$ with a flat optimum, removes this at a $\text{polylog}(T)$ cost (Zhang et al., 2026). *Unknown $\|u\|$ / unbounded domain:* Theorem 3.1 gives bounded feedback $\|\hat{g}_t\| \leq M$, exactly the input required by the parameter-free potential of Zhang & Cutkosky (2022); substituting our $\hat{g}_t$ yields the unbounded-domain analogue with D replaced by $\|u\|$. *Unknown regime length:* Theorem A.4 plus the interval version of Theorem A.3 give SN-OMD an interval-regret guarantee on every $[a, b]$; a strongly-adaptive wrapper (Daniely et al., 2015; Cutkosky, 2020) then recovers the dynamic bound without knowing the regime structure, at a polylog overhead. A per-regime transition cost $O(DN \text{polylog } T^{1/p})$ is incurred while the scale tracker re-concentrates after each of the N regime changes; it is dominated when regimes are macroscopic.

A.7. Discharging Theorem 3.2: a bracket guarantee

Theorem 3.2 posits a predictable scale that brackets σ_t . We show it is not an oracle: under a mild model of the drift, the two-timescale envelope of Section 4.2 *provably* meets the essential lower bracket off a small exceptional set, so the burden moves from “an accurate scale estimate exists” to “the drift is piecewise slowly varying,” and the estimate is then *derived* from standard robust concentration.

Drift model. Partition $[T]$ into N regimes by change points $t_0 < \dots < t_N$. *Slow within-regime drift:* inside a regime the scale varies slowly, $\sigma_t / \sigma_{t-1} \in [1 - \gamma, 1 + \gamma]$ with $\gamma W \leq c_0$ for the window W . *Arbitrary jumps:* at the N change points σ_t may jump by any factor. This is the piecewise-slowly-varying picture the measured gradient scale fits ($\sim 30\%$ within-day drift, occasional $3\times$ overnight jumps; Section 4).

Tracker. The envelope $s_t = \max\{c\hat{m}_t, (1 - \rho)s_{t-1}\}$, where $\hat{m}_t = \text{median}(\|g_{t-W}\|, \dots, \|g_{t-1}\|)$ is the predictable short-window median and $\rho \asymp 1/L$ with $L = \min_r L_r$.

Proposition A.5 (Bracket guarantee). *Under Theorem 2.1 and the drift model above, for a large enough constant c , with probability $\geq 1 - \delta$: (i, lower bracket) $s_t \geq c_1\sigma_t$ for every t outside at most $O(W)$ rounds after each jump—an exceptional set of size $O(NW)$; and (ii, bounded variation) $W_s = O(\sup_t \sigma_t + \rho \sum_t \sigma_t) = O(V_\sigma^+)$. Consequently Theorem 3.3 holds with one added term $O(DNW \sup_t \sigma_t)$, dominated whenever $NW \ll T^{1/p}$ (macroscopic regimes).*

Proof sketch. Window concentration. On a window lying inside one regime, the population median of $\|g_t\|$ is a constant multiple of σ_t (regular variation), and the empirical median concentrates around it by order-statistic bounds (needing essentially no moment); the slow drift shifts the constants by $O(\gamma W) = O(1)$. Hence $\hat{m}_t \in [a\sigma_t, b\sigma_t]$ w.h.p., and $c \geq c_1/a$ gives $c\hat{m}_t \geq c_1\sigma_t$, so the max meets the lower bracket whenever the window lies in one regime. *Jumps.* After an upward jump the window still holds pre-jump norms for $\leq W$ rounds; during that lapse the median term may under-estimate, giving the $O(NW)$ exceptional set. A downward jump only makes s_t conservatively high (the release lets it fall at rate ρ), never violating the lower bracket. *Variation.* Upward moves of s_t come only from $c\hat{m}_t$ rising; between jumps these telescope to $\sup_t s_t$, and the geometric release contributes the churn $\rho \sum_t s_t$, so $W_s \leq \sup_t s_t + \rho \sum_t s_t$, which is $O(V_\sigma^+)$ for $\rho \asymp 1/L$ (Theorem 3.5). *Exceptional rounds.* On the $O(NW)$ lapse rounds the update is still bounded ($\|\hat{g}_t\| \leq M$, Theorem 3.1) but the clipping bias (C) is uncontrolled; charging it trivially at $O(D \sup_t \sigma_t)$ per round gives the added $O(DNW \sup_t \sigma_t)$. $\square$

The residual—the post-jump lapse—is an additive term already of the form isolated in the Reductions above, and making ρ adaptive to an unknown L is exactly the strongly-adaptive wrapper of that subsection.

A.8. The deployed EMA vs. the analyzed envelope tracker

The regret analysis discharges Theorem 3.2 with the two-timescale “peak-hold” envelope of Section 4.2, whereas the empirically strongest configuration uses a winsorized exponential moving average (EMA). Because a reviewer may reasonably ask whether the analyzed object and the deployed object are the same, we state precisely what does and does not depend on the choice of tracker.

Stability is tracker-independent. Theorem 3.1 uses only $\|\hat{g}_t\| \leq M$, which the cap enforces for any predictable $s_t > 0$. Non-divergence therefore holds verbatim for the EMA, for the envelope, and for any other tracker; the gap discussed here is never about stability, only about the leading regret constant.

Only the regret constant depends on the tracker. For any predictable s_t meeting the essential lower bracket $s_t \geq c_1\sigma_t$, the rate is $T^{1/p}$ (Theorem 3.3); trackers differ only through the leading factor $W_s^{1-1/p}$. The envelope is the tracker for which we can prove both the lower bracket and $W_s = O(V_\sigma^+)$ (Theorem A.5)—reacting up on a short robust median and releasing down only geometrically keeps every upward move a genuine rise (Table 3: 100% lower bracket, $W_s = 6.7 V_\sigma^+$). A fast, *unwinsorized* EMA can in principle churn on heavy-tailed $\|g_t\|$ spikes (Table 3: $W_s = 62 V_\sigma^+$); winsorizing the EMA at $k \cdot \text{median}$ caps each upward step and is what keeps the deployed default’s W_s finite in practice, though we do not give it a closed-form bound.

Why deploy the EMA nonetheless. It is simpler, standard, and empirically more accurate—weighted $R^2 \approx 0.24$ versus ≈ 0.19 for the envelope on the continuous stream—because the envelope’s benign over-estimation of the scale shrinks its steps η/s_t into a mild underfit, which is precisely the price of its provable W_s . The two trackers are thus endpoints of a single accuracy/guarantee trade-off, and *both* are bounded; deployment can pick either without risking divergence.

Closing the gap. Either run the envelope in deployment, accepting the small accuracy cost, or prove a W_s bound for the winsorized EMA under a heavy-tailed scale process: the winsorization already supplies the per-step cap the envelope gets from its median, so what remains is to control the geometric-decay churn term $\rho \sum_t s_t$, exactly as in Theorem 3.5, with ρ the EMA rate. The one residual modeling assumption— ρ matched to an unknown regime length—is removed by the strongly-adaptive wrapper of the preceding subsection.